

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Level

MARK SCHEME for the June 2005 question paper

9702 PHYSICS

9702/04

Paper 4 (Core), maximum raw mark 60

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. This shows the basis on which Examiners were initially instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began. Any substantial changes to the mark scheme that arose from these discussions will be recorded in the published *Report on the Examination*.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the *Report on the Examination*.

- CIE will not enter into discussion or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the June 2005 question papers for most IGCSE and GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Grade thresholds for Syllabus 9702 (Physics) in the June 2005 examination.

	maximum mark available	minimum mark required for grade:		
		A	B	E
Component 4	60	41	35	19

The thresholds (minimum marks) for Grades C and D are normally set by dividing the mark range between the B and the E thresholds into three. For example, if the difference between the B and the E threshold is 24 marks, the C threshold is set 8 marks below the B threshold and the D threshold is set another 8 marks down. If dividing the interval by three results in a fraction of a mark, then the threshold is normally rounded down.



June 2005

GCE A LEVEL

MARK SCHEME

MAXIMUM MARK: 60

SYLLABUS/COMPONENT: 9702/04

PHYSICS
Paper 4 (Core)



UNIVERSITY of CAMBRIDGE
International Examinations

Page 1	Mark Scheme	Syllabus	Paper
	A LEVEL - JUNE 2005	9702	4

- 1 (a) (i) angular speed = $2\pi/T$ C1
 $= 2\pi/(3.2 \times 10^7)$
 $= 1.96 \times 10^{-7} \text{ rad s}^{-1}$ A1 [2]
- (ii) force = $mr\omega^2$ or force = mv^2/r and $v = r\omega$ C1
 $= 6.0 \times 10^{24} \times 1.5 \times 10^{11} \times (1.96 \times 10^{-7})^2$
 $= 3.46 \times 10^{22} \text{ N}$ A1 [2]
- (b) (i) gravitation/gravity/gravitational field (strength) B1 [1]
- (ii) $F = GMm/x^2$ or $GM = r^3\omega^2$ C1
 $3.46 \times 10^{22} = (6.67 \times 10^{-11} \times M \times 6.0 \times 10^{24})/(1.5 \times 10^{11})^2$ C1
 $M = 1.95 \times 10^{30} \text{ kg}$ A1 [3]
- 2 (a) obeys the law $pV/T = \text{constant}$ or any two named gas laws M1
at all values of p , V and T A1 [2]
or two correct assumptions of kinetic theory of ideal gas (B1)
third correct assumption (B1)
- (b) (i) mean square speed B1 [1]
- (ii) mean kinetic energy = $\frac{1}{2}m\langle c^2 \rangle$ M1
 $\rho = Nm/V$ and algebra leading to [do not allow if takes $N = 1$] M1
 $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$ A0 [2]
- (c) (i) $\frac{1}{2} \times 6.6 \times 10^{-27} \times (1.1 \times 10^4)^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times T$ C1
 $T = 1.9 \times 10^4 \text{ K}$ A1 [2]
- (ii) Not all atoms have same speed/kinetic energy B1 [1]
- 3 (a) (thermal) energy/heat required to convert unit mass/1 kg of solid to liquid M1
with no change in temperature/at melting point A1 [2]
- (b) (i) energy required to warm ice = $24 \times 10^{-3} \times 2.1 \times 10^3 \times 15$ (= 756 J) C1
energy required to melt ice at 0°C = $24 \times 10^{-3} \times 330 \times 10^3$ (= 7920 J) C1
total energy = 8700 J A1 [3]
- (ii) energy lost by warm water = $200 \times 10^{-3} \times 4.2 \times 10^3 \times (28 - T)$ C1
 $200 \times 4.2 \times (28 - T) = 24 \times 4.2 \times T + 8676$ C1
 $T = 16^\circ\text{C}$ A1 [3]
[allow 2 marks if ΔT calculated]
[allow 2 marks if $(24 \times 4.2 \times T)$ omitted]
[allow 1 mark for $224 \times 4.2 \times (28 - T) = 8676$, $T = 19^\circ\text{C}$]

Page 2	Mark Scheme	Syllabus	Paper
	A LEVEL - JUNE 2005	9702	4

- 4 (a) acceleration proportional to displacement (from a fixed point) M1
or $a = -\omega^2 x$ with a , ω and x explained A1 [2]
and directed towards a fixed point
or negative sign explained
- (b) for s.h.m., $a = (-)\omega^2 x$ B1
identifies ω^2 as $A\rho g/M$ and therefore s.h.m. (may be implied) B1
 $2\pi f = \omega$ B1
hence $f = \frac{1}{2\pi} \sqrt{\frac{A\rho g}{M}}$ A0 [3]
- (c) (i) $T = 0.60$ s or $f = 1.7$ Hz C1
 $0.60 = (2\pi\sqrt{M})/\sqrt{(\pi \times \{1.2 \times 10^{-2}\}^2 \times 950 \times 9.81)}$ C1
 $M = 0.0384$ kg A1 [3]
- (ii) decreasing peak height/amplitude B1 [1]
- 5 (a) field strength = potential gradient [- sign not required] B1 [1]
[allow $E = \Delta V/\Delta x$ but not $E = V/d$]
- (b) No field for $x < r$ B1
for $x > r$, curve in correct direction, not going to zero B1
discontinuity at $x = r$ (vertical line required) B1 [3]
- 6 (a) (i) flux/field in core must be changing M1
so that an e.m.f./current is induced in the secondary A1 [2]
- (ii) power = VI M1
output power is constant so if V_s increases, I_s decreases A1 [2]
- (b) (i) same shape and phase as I_p graph B1 [1]
- (ii) same frequency M1
correct phase w.r.t. Fig. 6.3 A1 [2]
- (iii) $\frac{1}{2}\pi$ rad or 90° B1 [1]
- 7 (a) curve levelling out (at $1.4 \mu\text{g}$) M1
correct shape judged by masses at $nT_{1/2}$ A1 [2]
[for second mark, values must be marked on y-axis]
- (b) (i) $N_0 = (1.4 \times 10^{-6} \times 6.02 \times 10^{23})/56$ C1
 $= 1.5 \times 10^{16}$ A1 [2]
- (ii) $A = \lambda N$ C1
 $\lambda = \ln 2/(2.6 \times 3600)$ ($= 7.4 \times 10^{-5} \text{ s}^{-1}$) C1
 $A = 1.11 \times 10^{12} \text{ Bq}$ A1 [3]
- (c) 1/10 of original mass of Manganese remains C1
 $0.10 = \exp(-\ln 2 \times t/2.6)$
 $t = 8.63$ hours A1 [2]
[use of 1/9, giving answer 8.24 hrs scores 1 mark]

Page 3	Mark Scheme	Syllabus	Paper
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- 8 (a) Q/V , with symbols explained [do not allow in terms of units] B1 [1]
- (b) (i) on a capacitor, there is charge separation/there are + and - charges M1
either to separate charges, work must be done
or energy released when charges 'come together' A1 [2]
- (ii) either energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
change = $\frac{1}{2} \times 1200 \times 10^{-6} (50^2 - 15^2)$ C1
change = 1.4 J (1.37) A1 [3]
[allow 2 marks for $\frac{1}{2}C(\Delta V)^2$, giving energy = 0.74 J]